

ECE 332 – WIRELESS POWER TRANSFER LAB

INTRODUCTION

In this lab, you will build and measure the performance of a basic wireless charger.

Note: You will not actually be charging a battery in this lab. Do not connect a battery to anything you build here; the additional circuitry required to recharge a Li-Po battery is not within the scope of this class. Li-Po batteries are volatile if mishandled and can explode if improperly charged.

This lab requires tools/equipment not found in the Emag Lab. You may need to use the Capstone Lab for some sections. You must solder all parts of this lab – don't use breadboards.

For lab reports, please answer all questions and provide all plots/screen captures requested in the Procedure sections. One lab report per group member, groups can be no larger than two people, and include both names on the lab report. Questions are not labelled (life isn't a textbook!); you'll have to read this document carefully to answer all of them. The basis of your grade for this lab will be as follows:

- 50% for following all directions – plots set up exactly as described, correct process being followed
- 25% for full participation – you completed every step and didn't skip sections, you asked questions about sections you didn't understand, and the report was submitted on time.
- 25% for getting correct answers within an acceptable margin of error.

REQUIRED HARDWARE

- Student Supplied: Lab Kit
 - Insulated wire, 22-28 AWG thickness, two 5-foot sections
 - Variable Capacitors (x2)
 - Schottky diodes (1N5819) x2
 - 4.1V Zener diode
 - Blue LED ($V_f \sim 2.8V$) x1
 - Side-launch SMA connector x2
 - BNC-to-SMA adapter kit (reusable)
 - Two ceramic (*not* electrolytic!) capacitors – any value from 100nF to 1.0 uF, but they should both be the same value – the lab kit has two 1.0uF caps in it
 - BNC cable from General ECE Lab Kit
- Lab Provided: VNA (located in the emag lab)
 - a NanoVNA is great if you have one, otherwise use the emag lab.
 - Calibration Kit for VNA
 - If you're in the emag lab you can use the eCal module (be gentle!), or you can use a mechanical standards kit. If you've never used the eCal module before, ask the TA.
 - If you don't know how to calibrate a VNA with the mechanical standards kit, ask the TA.
- Soldering iron & peripherals
 - Wire strippers
 - Small flathead screwdriver (or something similar)

PROCEDURE

VNA SET-UP

1. Turn on the VNA if it's not on already. Don't calibrate it yet.
2. Connect the eCal module via USB cable to any of the VNAs in the room and let it warm up.
 - a. The light on the eCal module will turn green when it's ready to use.
 - b. You'll all be sharing the same eCal module, leave it plugged into USB when not in use, so it stays warmed up.
3. Format the VNA to display the data we want to measure.
 - a. The 'Display' button will let you split the screen however you want. We only need one plot on the screen for now, so you can leave this alone.
 - b. The 'Format' button lets you choose between Smith/Log Mag/VSWR/etc. formats
 - c. The 'Meas' button sets which S-parameter we want to measure
 - d. Display a Log-Mag format (R+jX) measuring S₁₁ magnitude**
 - e. Display a Log-Mag format (R+jX) measuring S₁₁ phase**
 - i. These two plots should be overlapped on the screen
 - ii. The VNAs have touch-screens. You may have to tap on a plot to select it before changing its settings.
 - f. Set the Start frequency to 1.0 MHz and the Stop frequency to 20.0 MHz
 - g. Connect an SMA cable (the blue ones from the emag lab) to port 1 on the VNA.
4. Calibrate the VNA
 - a. Connect the port 1 cable to the eCal module
 - b. Select eCal after pressing the 'Cal' button, then follow the instructions.
 - c. Remember, if you change the start, stop, or center frequencies on the VNA at any point you'll have to recalibrate, so leave those alone.
 - d. If you unplug or change the port cables from the VNA, you'll also have to recalibrate before taking a measurement.
 - e. Again, if you don't remember how to calibrate a VNA, ask the TA. It's better to ask a question for the 37th time than to break a \$60,000 piece of equipment. The same is especially true for the eCal module, since PSU only has one of them.

BUILD AND TUNE THE TRANSMITTER COIL

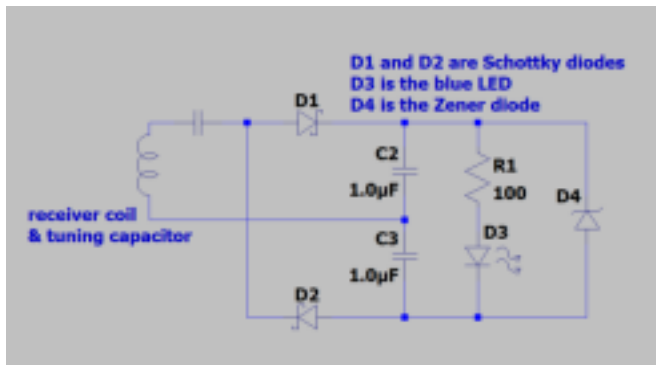
1. Using one of the 5' long pieces of wire, make a coil that is 3" in diameter with 6 turns. This should use most of the wire.
 - a. Tape the turns tightly together so they don't move around.
 - b. Don't tape the coil to the table or furniture, you'll be carrying it between labs.
2. Solder one leg of the variable capacitor to one end of the coil.
 - a. It doesn't matter which leg or which wire of the coil you choose.
 - b. Note: the variable capacitors we use here have three legs (for mechanical reasons), but two of them are connected together. It should be obvious to see which two are the same pin.
3. Solder the other pin of the capacitor to one of the outside legs of the SMA connector – it doesn't matter which, just not the center pin.
4. Solder the unused wire from the coil to the center pin of the SMA connector.
5. Your receiver coil is built – now it's time to tune it.

6. Leave the SMA connector attached to the transmitter coil.
7. Connect the coil to the VNA and observe the S_{11} plots.
 - a. The frequency where the coil is resonant should show a dip in the S_{11} magnitude plot. b. This dip may be small and hard to find the exact center, so if we look at phase, we see there's an asymptote in that plot at the resonant frequency. This is much easier to use for tuning.
8. Set a marker at 6.780 MHz on the VNA.
9. Slowly and carefully (i.e. don't break your solder joints) turn the screw on the variable capacitor until the phase asymptote is at an integer multiple of 6.78 MHz (6.78 MHz, 13.56 MHz, etc).
 - a. Note: when you touch the capacitor with your fingers you should notice the resonant frequency shift.
 - b. Remember, you are a sentient skin-bag of saline (a conductor) and your skin is a dielectric, so you are adding capacitance simply by existing near the device.
 - c. **You should only accept measurements of the tuning when you are not touching the coil at all.**
10. If you've followed all the instructions, you now have a transmitter coil tuned to 6.78 MHz (or an integer multiple of it) when you are not touching it.

BUILD AND TUNE THE RECEIVER COIL & RECTIFIER

1. Repeat the instructions for the transmitter coil to build the receiver coil, except:
 - a. The coil of wire for the receiver should be 12 turns with a 1" diameter. You will have more wire left over than you did from the previous coil – it can be trimmed off.
 - b. After both coils are tuned to the same frequency on the VNA, unsolder the SMA connector from the receiver coil.
 - i. Save the SMA connector, it can be reused for other projects.
 - ii. Be careful not to adjust the capacitor further or you'll have to reattach the SMA jack and retune with the VNA.
2. Add the circuit shown to the right to your tuned receiver coil; take note of diode orientation.
3. Don't worry about making the circuit look pretty as you build it, focus on making sure that nothing is shorting together where it shouldn't be.

- a. You should be able to carry this circuit around without anything shorting together.
- b. Keeping component leads trimmed short will help with durability and portability.



4. You now have a tuned receiver coil connected to a voltage-doubling rectifier circuit, which provides a DC voltage to light an LED. The Zener diode provides over-volt protection so you don't blow your LED out.

WIRELESS POWER TRANSFER

1. By now you should have a transmitter and receiver coil, both tuned to the same 6.78 MHz or integer

multiple frequency. If this is the case, and if you don't accidentally adjust the tuning capacitors, you are done in the emag lab, but there's more to do for this lab – don't leave for the day yet.

- a. Check the 'Deliverables' section of this document so you can get all the VNA screenshots before you leave the Emag Lab.
2. Go to the Capstone Lab (60-04) or the Tektronics Lab (60-01) – you need the signal generators. 3. Your PSU Lab Kit should have a BNC cable in it. If not, borrow/buy one from the EPL. 4. Connect the BNC cable to the output of a signal generator. Set a sine wave with the frequency of your coils (6.78 MHz or an integer multiple) and the amplitude at 3 V.
 - a. Connect the other end of the BNC to a BNC-to-SMA adapter, then connect that to your transmitter coil.
5. Hold the receiver coil parallel to the transmitter coil. If the LED doesn't light up, move it closer. a. You may have to turn the voltage up on the signal generator – document the amplitude you use. 6. Document how far apart the coils are where you can still observe the LED turning on. 7. Change to a square wave, repeat the previous two steps.
8. What happens if you turn the receiver coil orthogonal to the transmitter coil? What if you bring it close to the edge of a coil? Explain in terms of flux and magnetic field lines.
9. Does it work if your lab partner puts their hand in between the transmitter and receiver? Are you surprised by this outcome? Why or why not?
10. How could you improve the efficiency of this? Explain in detail (like you'll be graded on it).

DELIVERABLES FOR LAB REPORT

1. Images
 - a. Screen captures of VNA for both coils showing tuning plots – one screen capture per coil.
 - b. Photo(s) of both coils laying on a flat surface. A ruler should be in the photo to establish scale.
 - c. Photo(s) of the LED being lit from wireless power transfer.
 - d. Any interesting photos you'd like to include from this project.
2. Data
 - a. Answer all questions and discussion prompts found in the lab procedure.
 - b. Transmitter amplitude
 - c. Frequency adjustments (if you didn't use 6.78 MHz that's fine, but explain why).
 - d. Maximum coupled distance where LED was still perceptibly on
3. Structure and Organization
 - a. Format and organize the Images and Data into a lab report to include all group members' names, images, and plots.
 - b. Formatting requirements to be set by lab TA.